

Positive Definiteness and Carryover Sensitivity in the Hendrickson et al. (2020) Simulation Framework

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Purpose. The simulation framework of Hendrickson et al. [1] has two known problems. First, many of the correlation matrices its data-generating process (DGP) builds are not positive definite, and the code forces them to positive definiteness by a step the paper does not describe. Second, the power of the Hybrid N-of-1 design is extremely sensitive to even very small carryover. This paper documents both problems from the published code and results, identifies a common cause, and evaluates three adjustments proposed to address them: (1) replacing compound symmetry (CS) with AR(1) within-factor correlation, (2) letting the biomarker-response correlation decay off drug, and (3) carrying the biomarker-treatment interaction in the mean of the DGP.

1. Summary

1. **Problem 1 is confined to the biomarker row and is systematic.** Running the published `generateData()` at the published parameters, 40 of the 162 design-path-cell matrices fail the code’s own positive definiteness check: every matrix with $c_{bm} > 0$ and no carryover in the CS, OL+BDC and Hybrid designs, plus the placebo-first CO path at every half-life. The response block itself is always valid. The failures occur because the requested biomarker-response correlation exceeds the largest value the CS structure can carry, 0.256 to 0.262 in these designs, well below the published values of 0.3 and 0.6 (Section 4).
2. **The undisclosed repair changes the simulated effect.** At $c_{bm} = 0.6$ the repair lowers the on-drug biomarker-response correlation to 0.54-0.57, introduces 0.03-0.04 at off-drug visits, and moves other correlations by up to 0.10. At $c_{bm} = 0.3$ the change is at most 0.007 (Section 4.3).
3. **Problem 2 is an artifact of the coupling rule, not of carryover.** In the published results, Hybrid power at $N = 70$, $c_{bm} = 0.3$ falls from 0.74 with no carryover to 0.12 at a half-life of 0.1 weeks, about 17 hours, when the residual drug effect one week after stopping is 0.1% of its on-drug value. The published rule sets the biomarker-response correlation to its full value wherever any residual effect remains, so with any carryover the coupling is constant across visits, the on/off contrast that identifies the interaction disappears, and the estimate collapses toward zero (-0.28 to -0.07 at $c_{bm} = 0.6$) (Section 5).
4. **The two problems share one cause.** Under the step rule, a cell without carryover has the contrast but an invalid matrix, and a cell with any carryover has a valid matrix but no contrast. The “sensitivity to small carryover” is the jump between those two regimes at a half-life of zero (Section 5.4).
5. **The three adjustments are not independent.** Decaying coupling (adjustment 2) removes the jump, but under CS it leaves the ceiling at 0.26 in every cell, so applied alone it converts problem 2 into problem 1 everywhere. AR(1) (adjustment 1) raises the ceiling but does not remove the jump. Together, with the separable AR(1) structure of `docs/35`, the Hybrid ceiling is 0.44 at the published $\rho = 0.8$: enough for $c_{bm} = 0.3$, not for 0.6. Mean moderation (adjustment 3) removes the ceiling altogether, but it is a different model of the biomarker rather than a repair of the covariance construct (Section 6).

2. Evidence basis

Evidence labels: **verified** (computed in this work), **inspected** (read in code or documents), **inferred** (follows from verified facts), **unverified**.

- **Published code and results.** All statements about the published framework refer to `github.com/rchendrickson/pmsimstats` at commit 58b32a9 (19 November 2020), the last commit before publication, here called `orig`. The repository at that commit contains the publication vignettes and the saved replicate-level results (`results_core`: 4 designs, 18 model cells, 5 censoring conditions, 1,000 replicates per cell). The data-generating code at 58b32a9 is identical to the initial commit 42ac030 (inspected, git history).

- **Problem 1** was established by running the published `generateData()` itself with the published parameters, capturing the covariance matrix passed to `mvrnorm`, and applying the published check and repair (verified). An independent reconstruction of the matrices from the specification gave identical failure patterns (verified).
- **Problem 2** was established from the published saved results, with no re-simulation (verified).
- **Adjustments** were evaluated by exact positive-definiteness ceilings at the published parameters (verified). No power simulation under the adjustments has been run yet (Section 9).

3. The published framework

Each participant's outcome is $Y_{it} = BL_i - (TV_{it} + PB_{it} + BR_{it})$, with the biomarker B_i , baseline and the three response components drawn jointly from a multivariate normal distribution. The parameters used for the published results were (inspected, `Produce_Publication_Results_1_generate_data.Rmd`, and the saved `parameterselections`):

Parameter	Published values
Designs	OL, OL+BDC, CO, Hybrid ("N-of-1"); 8 visits over 20 weeks
N	35, 70
c_{bm}	0, 0.3, 0.6
Carryover half-life $t_{1/2}$	0, 0.1, 0.2 weeks
Within-factor correlation ρ	0.8 for all three components, constant at every lag (CS)
Cross-factor correlation	$c_1 = 0.2$ same visit, $c_{\times} = 0.1$ otherwise
Biomarker-BR correlation	c_{bm} wherever the carryover-adjusted BR mean is nonzero, 0 elsewhere
Carryover on BR mean	recursive, $\mu_{p-1}(1/2)^{t_{sd}/t_{1/2}}$ with cumulative t_{sd}
Positive definiteness	<code>makePositiveDefinite = TRUE</code> in every call: <code>corpcor::make.positive.definite(sigma, tol = 1e-3)</code> when the check fails
Analysis	<code>lmer</code> , random intercept, binary on-drug indicator, test of <code>bm:Db</code>

The Hybrid schedule places visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20; all four of its paths begin on drug, as do both OL+BDC paths. CO has one path on drug first and one placebo first.

4. Problem 1: correlation matrices that are not positive definite

4.1 Incidence

Failures of the published check, by design (verified):

Design	Matrices checked	Failures	Where
OL	18	0	none
OL+BDC	36	8	both paths, $c_{bm} > 0$, $t_{1/2} = 0$
CO	36	16	both paths at $t_{1/2} = 0$; the placebo-first path at every $t_{1/2}$ ($c_{bm} > 0$)

Design	Matrices checked	Failures	Where
Hybrid	72	16	all four paths, $c_{bm} > 0$, $t_{1/2} = 0$
Total	162	40	

Every matrix with $c_{bm} = 0$ passes. Counts treat the two values of N separately, as in `docs/07`, although N does not enter the matrix.

4.2 Mechanism

The response block (all *TV*, *PB* and *BR* entries) is positive definite in every case: under CS with constant cross-factor terms its smallest eigenvalue is $(1 - \rho) - (c_1 - c_\times) = 0.1$, independent of the design (verified). Failure therefore always comes from the biomarker row. With $\tilde{r} = c_{bm}v$ the biomarker's correlations with the *BR* visits, the Schur complement gives the exact largest feasible value

$$c_{bm}^* = (v^\top M^{-1}v)^{-1/2},$$

where M is the response block and v the coupling pattern (`docs/34`, Section 3.2). For the published construct at the published parameters (verified):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.1$	$t_{1/2} = 0.2$
OL	0.893	0.893	0.893
OL+BDC	0.262	0.893	0.893
CO	0.256	0.256	0.256
Hybrid	0.256	0.893	0.893

Both published effect sizes, 0.3 and 0.6, exceed the ceiling wherever it is 0.26, which reproduces the failure pattern of Section 4.1 exactly. The ceiling is high when the coupling pattern is constant across visits (OL always; OL+BDC and Hybrid with any carryover, Section 5.3) and low when it has an on/off contrast (no carryover, or CO's placebo-first path, whose pre-exposure visits keep a zero coupling at every half-life).

4.3 What the repair does

When the check fails, the code replaces the covariance matrix by a nearby positive definite one, raising small and negative eigenvalues to a floor, in the spirit of nearest-positive-definite approximation [2]. The effect on the correlations actually simulated (verified):

c_{bm} specified	On-drug correlation used	Off-drug correlation used	Largest change elsewhere
0.3	0.298-0.300	0.000-0.001	0.007
0.6	0.540-0.568	0.029-0.042	0.097

At $c_{bm} = 0.3$ the repair is cosmetic. At $c_{bm} = 0.6$ the simulated effect is materially smaller than specified, some interaction signal is moved into off-drug visits, and the within- and cross-factor correlations are altered as well. The published no-carryover results at $c_{bm} = 0.6$ therefore describe a DGP other than the one stated.

5. Problem 2: sensitivity of Hybrid power to carryover

5.1 The published results

Power from the published saved results, without dropout (verified):

Design	N	c_{bm}	$t_{1/2} = 0$	0.1	0.2
Hybrid	35	0.3	0.54	0.08	0.07
Hybrid	35	0.6	0.95	0.18	0.23
Hybrid	70	0.3	0.74	0.12	0.10
Hybrid	70	0.6	1.00	0.22	0.33
OL+BDC	70	0.3	0.74	0.19	0.13
OL+BDC	70	0.6	0.99	0.36	0.41
CO	70	0.3	0.85	0.44	0.46
CO	70	0.6	1.00	0.90	0.91

Type I error at $c_{bm} = 0$ lies between 0.01 and 0.09 across cells.

5.2 The carryover involved is negligible

With a half-life of 0.1 weeks, the residual drug effect one week after discontinuation, the earliest off-drug visit in Hybrid and OL+BDC, is $2^{-10} \approx 0.001$ of its on-drug value; at 0.2 weeks it is $2^{-5} \approx 0.031$. A carryover this small cannot plausibly erase three quarters of the power through its effect on the response mean.

5.3 Mechanism: the step rule removes the contrast

The published rule assigns c_{bm} to every visit at which the carryover-adjusted BR mean is nonzero. Exponential and recursive carryover never reach exactly zero, so with any positive half-life every visit after first exposure receives the full c_{bm} . In Hybrid and OL+BDC, whose paths all begin on drug, the coupling becomes constant across all eight visits (verified, captured matrices: Hybrid path A coupling 0.6 at every visit for $t_{1/2} = 0.1$). The biomarker then predicts BR equally on and off drug, which is a biomarker main effect, and the biomarker-by-treatment interaction has no population counterpart.

The published estimates show the consequence directly. The mean of the interaction estimate in Hybrid at $N = 70$ (verified):

c_{bm}	$t_{1/2} = 0$	0.1	0.2
0.3	-0.151	-0.036	-0.036
0.6	-0.279	-0.065	-0.081

The estimate shrinks toward zero by roughly three quarters. The loss of power is primarily attenuation of the effect being estimated, not only inflation of its standard error. CO is affected less because its placebo-first path retains a zero coupling before first exposure, which preserves some contrast (and keeps that path's matrix invalid). OL has no off-drug visits, so its test does not rest on a drug contrast in the first place.

5.4 The common cause

	No carryover ($t_{1/2} = 0$)	Any carryover ($t_{1/2} > 0$)
Coupling pattern (Hybrid, OL+BDC)	on/off contrast	constant
Matrix at $c_{bm} = 0.3, 0.6$	invalid, repaired	valid
Interaction identified	yes	no

The step rule makes the coupling a discontinuous function of the half-life: any positive value, however small, flips the pattern from contrast to constant. Both problems are consequences of that one feature. Problem 1 is the price of the contrast at $t_{1/2} = 0$, and problem 2 is the loss of the contrast at $t_{1/2} > 0$.

6. The three adjustments

6.1 Adjustment 1: AR(1) instead of compound symmetry

AR(1) lets correlation fall with the time between visits. At the published $\rho = 0.8$ it raises the no-carryover ceiling modestly in Hybrid and OL+BDC (0.256 to 0.291; 0.262 to 0.303) and substantially in CO (0.256 to 0.627) (verified, Section 6.4). On its own it does not change the step rule, so with any carryover the coupling is still constant and the contrast is still lost. It addresses problem 1 in part and problem 2 not at all.

6.2 Adjustment 2: a correlation that decays off drug

Under the graded rule the off-drug coupling is $c_{bm}\phi_t$, with ϕ_t the residual fraction of the on-drug effect, so the coupling is a continuous function of the half-life. At $t_{1/2} = 0.1$ the residual at the first off-drug visit is 0.001, and the graded coupling is indistinguishable from the no-carryover pattern: the discontinuity of Section 5.4 is removed, and small carryover produces a small change. This addresses problem 2 directly.

Two qualifications. First, the published rule does not send the off-drug correlation to zero under carryover; it sends it to the full on-drug value. The graded rule sits between the published rule's two regimes, equal to zero-carryover behavior at negligible half-lives and decaying smoothly as the half-life grows. Second, because the graded pattern keeps the on/off contrast, it keeps the low ceiling: under CS it is 0.256-0.265 in every cell (verified). Applied alone, adjustment 2 would make $c_{bm} = 0.3$ and 0.6 infeasible in every Hybrid and OL+BDC cell, converting problem 2 into problem 1 everywhere. It must be combined with a structure that admits a larger ceiling.

6.3 Adjustment 3: the interaction in the mean

Under mean moderation the biomarker shifts the on-drug BR mean by $\beta_{bm}\sigma_{BR}b_i$ and has no correlation with BR in the matrix. The biomarker row is then trivially valid and there is no ceiling on the moderation parameter; only the response block must be positive definite, which it is under the published CS structure for every schedule (Section 4.2) and under the separable AR(1) structure of `docs/35` for every schedule (inferred from `docs/35`, Section 4.2). The interaction is carried by a mean shift that does not depend on the coupling rule, so the discontinuity of Section 5.4 does not arise. Adjustment 3 therefore addresses both problems.

It does so by changing the model rather than repairing it. Mean moderation asserts that the biomarker fixes each participant's drug effect; covariance moderation asserts that it predicts the effect only probabilistically, which is the rationale Hendrickson et al. gave for their construct [1] and the subject of paper 01 of this compendium. In paper 01's terms, adjustment 3 replaces Architecture B with Architecture A. It also introduces a known mismatch with an analysis model that uses a continuous exposure variable, because the mean shift is applied on drug only (paper 01, Section 2.2.1).

6.4 The adjustments in combination

Ceilings at the published parameters ($\rho = 0.8$, $c_1 = 0.2$, $c_x = 0.1$), design-level minimum over paths (verified):

Configuration	Hybrid, $t_{1/2} = 0 / 0.1 / 0.2$	OL+BDC	CO
Published (CS, constant cross, step)	0.256 / 0.893 / 0.893	0.262 / 0.893 / 0.893	0.256 / 0.256 / 0.256
Adjustment 1 only (AR(1), step)	0.291 / 0.595 / 0.595	0.303 / 0.594 / 0.594	0.627 / 0.576 / 0.576
Adjustment 2 only (CS, graded)	0.256 / 0.256 / 0.257	0.262 / 0.262 / 0.265	0.256 / 0.256 / 0.256
Adjustments 1 + 2, current covar	0.291 / 0.291 / 0.302	0.303 / 0.303 / 0.314	0.627 / 0.627 / 0.627
Adjustments 1 + 2, separable AR(1)	0.440 / 0.440 / 0.446	0.468 / 0.469 / 0.476	0.637 / 0.637 / 0.637
Adjustment 3 (mean moderation)	no ceiling	no ceiling	no ceiling

Read against the published effect sizes:

- $c_{bm} = 0.3$ is feasible in every cell only with adjustments 1 and 2 together in the separable form, or with adjustment 3. The current **covar** form of 1 + 2 is marginal in Hybrid (0.291 against 0.3).
- $c_{bm} = 0.6$ exceeds every covariance-moderation ceiling in Hybrid and OL+BDC at $\rho = 0.8$. No covariance construct examined can represent the published large effect in those designs; only mean moderation can. At the project's $\rho = 0.7$ the separable Hybrid ceiling is 0.48-0.49 (docs/35), still below 0.6.

6.5 Which adjustment addresses which problem

	Problem 1 (invalid matrices)	Problem 2 (carryover sensitivity)
1. AR(1)	Partly: raises the ceiling, CO most	No: the step remains
2. Graded coupling	No: alone it lowers the ceiling in carryover cells to the no-carryover level	Yes: removes the discontinuity
1 + 2 (separable)	Yes for $c_{bm} \leq 0.44$ (Hybrid, $\rho = 0.8$)	Yes
3. Mean moderation	Yes: no ceiling	Yes, by a different model

The evidence for the problem-2 entries is structural (the coupling pattern and the ceiling); the corresponding power has not yet been simulated (Section 9).

7. Implications

- **For the published results.** The no-carryover cells with $c_{bm} > 0$ in OL+BDC, CO and Hybrid, and the placebo-first CO path throughout, were generated from repaired matrices; at $c_{bm} = 0.6$ the simulated effect was smaller than stated (inferred from Section 4.3). The large loss of Hybrid and OL+BDC power with carryover is a property of the coupling rule, not of carryover, and should not be read as evidence that these designs are fragile to realistic carryover (inferred from Section 5).

- **For the project’s DGP.** Adopt adjustments 1 and 2 together, in the separable form, as the covariance-moderation DGP; compute the ceiling for every cell before simulating; and remove the silent repair, so that an infeasible effect size is an error rather than a quietly different simulation. Choose effect sizes below the ceiling, and report the ceiling alongside power.
- **On large effects.** If effect sizes near 0.6 must be studied in Hybrid or OL+BDC, the covariance construct cannot carry them, and the choice is between mean moderation (adjustment 3), with its different biological commitment, and a smaller within-factor correlation.

8. Relation to earlier project documents

- `docs/07` (positive definiteness failures) reports the same total of 40 failures in 162 checks, but its breakdown by design (32 of 40 in OL+BDC, 2 of 24 in Hybrid) and its explanation through the expectancy scaling of the placebo variance do not reproduce: positive diagonal scaling cannot change positive definiteness, and the published code’s own matrices fail as in Section 4.1. Superseded by Section 4.
- `docs/07a` gives an empirical ceiling of about 0.6 under an AR(1) structure the published code does not use. Superseded.
- `docs/09` (carryover correlation artifact) correctly identifies the step rule, but attributes the loss to standard-error inflation with an unbiased estimate; the published estimates attenuate by about three quarters (Section 5.3). It also contains arithmetic slips in the residual fractions.
- `docs/13` (Figure 4 walkthrough) labels its code as the initial commit but describes the 2024 revision’s coupling, scale factor and analysis variable.
- `docs/32` and `docs/34-docs/35` are consistent with this paper and supply the ceiling formula and the separable structure.

9. Open work

1. **Power under the adjustments.** Re-run the published grid (designs, N , c_{bm} , $t_{1/2}$ extended to 0.5 and 1.0) under: the published construct without repair (feasible cells only); adjustment 1; adjustments 1 + 2, separable; and adjustment 3. Use the published analysis model for comparability with [1], and the project’s continuous-exposure model alongside it.
2. **Separate repair from flattening.** Quantify how much of the published no-carryover power at $c_{bm} = 0.6$ reflects the repaired matrix, by simulating at c_{bm} values just below the ceiling with no repair.
3. **Hold other differences fixed.** The project DGP also corrects the recursive carryover and uses $\rho = 0.7$; the comparison in item 1 should vary one change at a time from the published configuration.
4. **Mean moderation’s analysis mismatch.** Evaluate adjustment 3 with both the binary and the continuous exposure variable.

10. Limitations

- The ceilings and failure counts are exact for the published designs and parameters; other designs and parameter values will differ.
- The repair was applied to the covariance matrix as in the published code, and its effect is reported on the correlation scale.
- Power under the adjustments is not yet simulated, so the problem-2 conclusions for the adjustments rest on the structure of the coupling and the ceiling, not on power.
- Reference [2] describes the class of method; the exact algorithm of `corpcor::make.positive.definite` was characterized here by its effect, not by source review.

11. Reproducibility

```
bash analysis/scripts/quick-sim/hendrickson-problems/00-fetch-58b32a9.sh
Rscript analysis/scripts/quick-sim/hendrickson-problems/01-published-matrices.R
```

```
Rscript analysis/scripts/quick-sim/hendrickson-problems/02-published-power.R
Rscript analysis/scripts/quick-sim/hendrickson-problems/03-adjustment-ceilings.R
```

Run from the repository root. The first script fetches the 58b32a9 sources and saved results with the GitHub CLI into `analysis/data/quick-sim/hendrickson-58b32a9/`; the others write their tables to `analysis/data/quick-sim/hendrickson-problems/`.

12. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020;2. Code: github.com/rchendrickson/pmsimstats, commit 58b32a9.
2. Higham NJ. Computing a nearest symmetric positive semidefinite matrix. *Linear Algebra and its Applications* 1988;103:103-118. (Cited from bibliographic knowledge; to be verified.)